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Sarcopenia risk in U.S. younger adults: the impact of physical activity intensity and occupational engagement—insights from a cross-sectional NHANES study

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Abstract

Objective This study aimed to investigate the associations between physical activity (PA) intensity, occupational PA engagement, and sarcopenia in younger adults.

Methods Data from the National Health and Nutrition Examination Survey (NHANES) (from 2011 to 2018) were used for the study. PA data were extracted for the Global Physical Activity Questionnaire, and the ratio of limb muscle mass (ALM) to body mass index (BMI) was used to determine sarcopenia. Multiple logistic regression was performed, and dose-response curves were drawn to evaluate the associations.

Results Of the 8802 adults aged 20 to 59 years, 770 (8.7%) were identified as having sarcopenia. Compared to those in the reference group, the participants in the “low”, “moderate”, and “high” total METs groups had lower risks of sarcopenia (low: OR = 0.730, 95% CI = 0.583; 0.915; moderate: OR = 0.643, 95% CI = 0.485; 0.853; high: OR = 0.560, 95% CI = 0.457; 0.687). The risk of sarcopenia decreased in the high-intensity recreational PA group; however, such a significant association was observed only in the high-intensity occupational PA group. The relationship between total METs and sarcopenia showed a nonlinear U-shaped trend.

Conclusions A greater intensity of PA is associated with a lower risk of sarcopenia among younger adults, regardless of whether moderate or vigorous PA is performed. Moreover, recreational PA may particularly advantageously contrast with occupational PA in lowering the risk of sarcopenia.

Keywords Younger adults, Physical activity, Intensity, Sarcopenia, NHANES

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Introduction

Sarcopenia, a condition characterized by a progressive decline in muscle mass, strength, and function that typically begins in adulthood, is a prevalent global health concern [1–3]. The reported prevalence of sarcopenia ranges from 8 to 36% worldwide, depending on the specific criteria utilized in research studies [4]. Current standards for diagnosing sarcopenia involve a comprehensive assessment of muscle mass, strength, and function, as outlined by organizations such as the European Working Group on Sarcopenia in Older People (EWGSOP) [5, 6] and the Asian Working Group for Sarcopenia (AWGS) [7], as well as definitions based on single measurement such as muscle mass [8] and handgrip strength [9], as low muscle mass or strength are related to functional decline [10, 11]. The detrimental consequences of sarcopenia include an elevated risk of falls, fractures, physical impairment, and mortality accompanied by a large disease burden [12–15], making sarcopenia a serious public health issue that needs urgent attention [16–19]. Aging is commonly acknowledged as a significant risk factor for sarcopenia, yet it is important to recognize that there are various other risk factors that may contribute to the development of sarcopenia in early adulthood, aside from age [20].

Prevention has been considered the main strategy for reducing the incidence of sarcopenia due to the current lack of effective treatment. Many researchers believe that regular physical activity (PA), a component of a healthy lifestyle, may have the potential to prevent sarcopenia [21, 22]. PA is defined as any bodily movement that is generated by skeletal muscles and requires energy expenditure, including activity in occupation, transportation, and leisure pursuits [23]. The literature has revealed that PA intensity might be an important issue in sarcopenia risk reduction; i.e., PA at a higher intensity strongly reduces sarcopenia risk compared with PA at a lower intensity [14, 24–26]. Recent literature highlighted the importance of the occupational nature of PA, i.e., despite the general health benefits of PA, occupational PA seems to increase the risk of adverse health outcomes [27–30], indicating the importance of differentiating occupational nature in developing precise strategies for disease prevention.

Although the relationship between PA and sarcopenia has been studied widely, research on sarcopenia in younger adults (e.g., before geriatric age) is very rare, although the etiologies of sarcopenia are highly prevalent among younger adults [20]. Additionally, evidence regarding the effect of the occupational nature of PA on sarcopenia has not been reported. Since the development of sarcopenia is related to many factors, such as nutritional metabolism, lifestyle and the environment, which may vary across different socioeconomic statuses [16, 31], the occupational nature of PA may have different

impacts on sarcopenia. However, the existing evidence addressing this issue remains very rare.

Therefore, the current study aimed to investigate the associations between the intensity and occupational nature of PA and the risk of sarcopenia. Specifically, this study may answer the following questions: (1) Are there dose-response relationships between PA intensity and sarcopenia? Does vigorous and moderate PA have a similar influence on sarcopenia? (2) Does occupational PA have a similar influence on sarcopenia compared with entertainment PA? The findings of the current study may provide clues for developing more precise PA recommendations for adults, thus holding great scientific and practical importance for the early prevention of sarcopenia.

Method

Study design and data collection

The data for the current study were obtained from the National Health and Nutrition Examination Survey (NHANES), which is an ongoing study with a multi-stage design that began in the early 1960s. The NHANES aimed to evaluate the health and nutritional status of Americans, and the findings could be used to determine the prevalence of major diseases and their risk factors. The surveys were conducted every 2 years, and a complex probability sampling method was used to represent the noninstitutionalized population in the United States. Data on sociodemographic characteristics, lifestyle factors, disease history, and various anthropometric measures of the participants were collected by trained interviewers and physicians. The questionnaire we used for the data collection in the current study has been published elsewhere [23]. The study protocols were approved by the National Center for Health Statistics Research Ethics Review Board, and all participants provided informed consent.

In the present study, data from 4 cycles of the NHANES (2011~2012, 2013~2014, 2015~2016, and 2017~2018) were used, with an initial sample of 39,156 participants. Participants who met the following criteria were excluded: (1) were aged younger than 20 years ($n=16538$); (2) had any missing data on dual-energy X-ray absorptiometry and body mass index ($n=11792$); (3) had any missing PA information ($n=54$); and (4) had any missing data on other variables that were necessary for data analysis, including smoking status, alcohol consumption, depression, hypertension status, diabetes status, hyperlipidemia status, and the household poverty-to-income ratio ($n=1970$). Ultimately, 8802 participants were included in the data analysis in this study (Fig. 1).

Assessment of PA

The PA of the participants was assessed using the NHANES questionnaire, which was designed based on

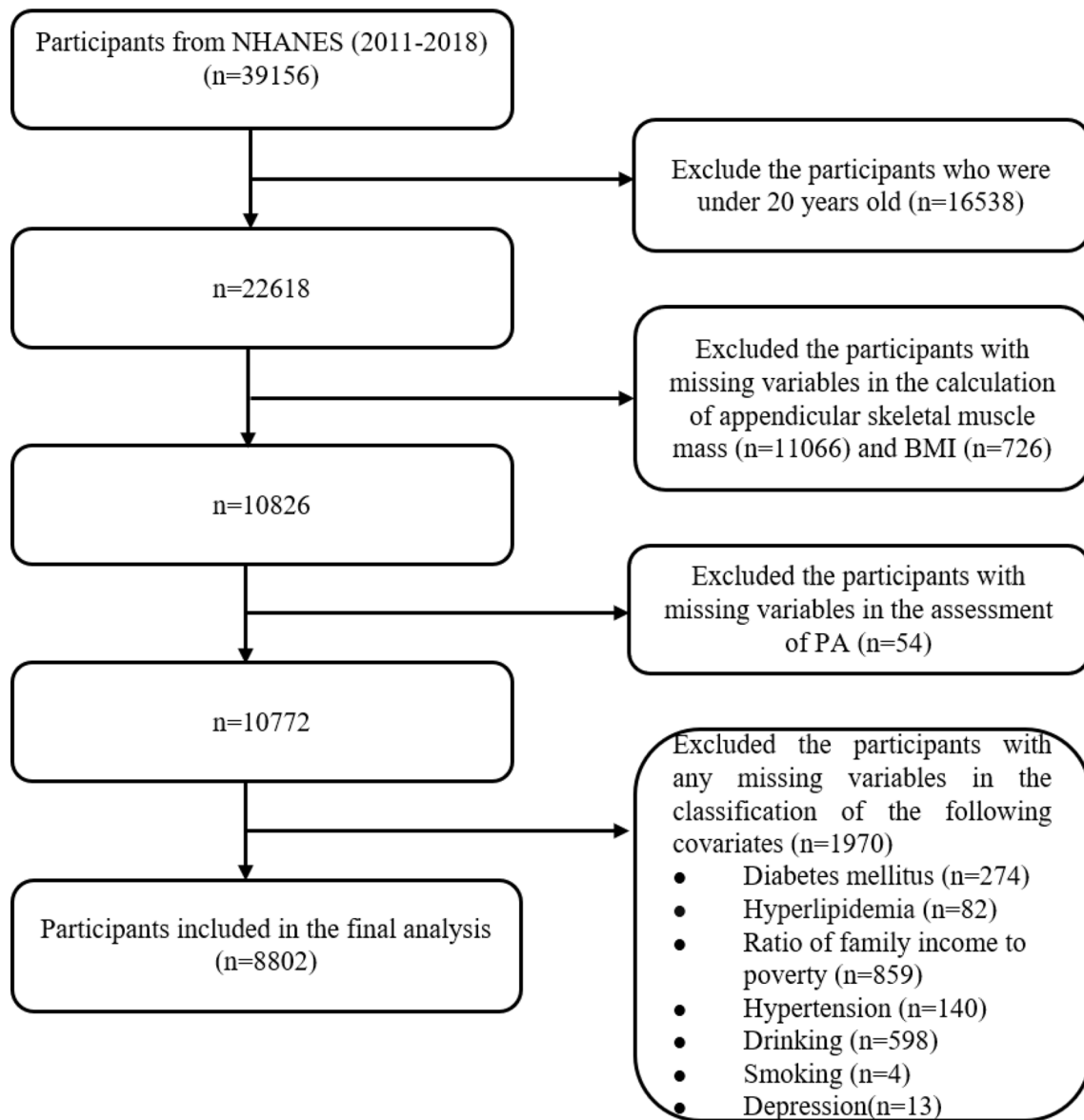


Fig. 1 Flowchart of participant selection

the Global Physical Activity Questionnaire (GPAQ) [23]. The frequency (times in a typical week), duration (minutes spent), and intensity (vigorous or moderate) of PA in three domains (occupational, transportation, and recreational) during a typical week were assessed, and the metabolic equivalent (MET) in a typical week was calculated by the following formula [14]:

$\text{MET} = \text{activity equivalent score (recommended by NHANES [32])} \times \text{frequency (active days per week)} \times \text{duration (minutes spent on an active day)}$.

METs of moderate activity were defined as the sum of the moderate METs of occupation, cycling, and recreation activities, and the METs of vigorous activity were defined as the sum of the vigorous METs in occupation and recreation activities. The number of METs involved

in moderate and vigorous activities was summed to obtain the total number of METs. The MET data were divided into four categories (none: 0, low: 1-1000, moderate: 1001-2000, high: >2001) according to the skewed data distribution.

To test the associations between the intensity of PAs with different occupational natures (i.e., occupational and recreational) and sarcopenia, questionnaire-assessed PAs during occupational were categorized as occupational, and questionnaire-assessed PAs during recreational activities were categorized as recreational. Similarly, the METs of moderate activity and vigorous activity with different occupational natures were also defined.

Definition of Sarcopenia

Dual-energy X-ray absorptiometry (DXA) scans were performed to obtain limb muscle mass data, and the masses of the four limbs were combined to obtain the appendicular limb muscle mass (ALM). According to the National Institutes of Health Foundation, sarcopenia is determined by the ratio of the ALM to body mass index (BMI), and male participants with an ALM with a BMI less than 0.789 kg/m² or a female participant with an ALM with a BMI less than 0.512 kg/m² were considered to have sarcopenia in the present study [8].

Covariates

The sociodemographic characteristics, lifestyle factors, and disease history of the participants were considered covariates in the study. Sociodemographic information included age, sex, marital status (married or unmarried), educational attainment (below high school, above high school), race (Mexican-American, non-Hispanic white, non-Hispanic black, and other races), and socioeconomic status (non-poor: a poverty-to-income ratio equal to one or higher, poor: a poverty-to-income ratio lower than one) [33]. Lifestyle factors included smoking status (non-smoking and smoking), alcohol consumption (not drinking and drinking), as any smoking or alcohol consumption is harmful [34, 35]. Nutritional status was classified into three categories (normal: BMI < 25.0, overweight: BMI = 25.0–29.9, obesity: BMI > 29.9) according to the WHO criteria [8, 36]. However, underweight participants (BMI < 18.5) were combined with those who had a normal weight because of the small proportion of underweight participants in the sample (1.6%). Patients with a history of hypertension, diabetes mellitus, or hyperlipidemia were classified into dichotomous groups (yes, no). Hypertension was determined if the participants had any of the following conditions: (1) systolic blood pressure ≥ 140 mmHg; (2) diastolic blood pressure ≥ 90 mmHg; or (3) had been formally diagnosed [33]. Diabetes mellitus was defined if the participants had any of the following conditions: (1) had been formally diagnosed by a physician or were taking anti-diabetes medication (pills or insulin); (2) had a glycemic hemoglobin (HbA1c) ≥ 6.5%; or (3) had a fasting blood glucose ≥ 126 mg/dL [37]. Hyperlipidemia was determined if the participants had any of the following conditions: (1) had been formally diagnosed; (2) had total cholesterol ≥ 200 mg/dL; (3) had triglycerides ≥ 150 mg/dL; (4) had high-density lipoprotein < 40 mg/dL (for males) or < 50 mg/dL (for females); or (5) had low-density lipoprotein ≥ 130 mg/dL [38]. Depression was measured using the Patient Health Questionnaire-9 (PHQ-9), a nine-item instrument that assesses the frequency of depressive symptoms over the past 2 weeks. The continuous total

PHQ-9 score (ranging from 0 to 27) was introduced into the regression model [23, 39].

Statistical analysis

The characteristics of the study population are expressed as the means and standard deviations (SDs) for continuous variables and percentages for categorical variables. For component differences for continuous variables, a *t* test or ANOVA was used. The chi-square test was used for between-group differences in categorical variables. Binary logistic regression models were used to estimate the association between PA intensity and sarcopenia, and odds ratios (ORs) and corresponding 95% confidence intervals (CIs) were used to describe these associations. Three models were developed to test the associations: Model 1 was non-adjusted; Model 2 was adjusted only for sex and age; Model 3 was adjusted for race, education, marital status, poverty-to-income ratio, nutritional status, smoking, alcohol consumption, depression, hypertension, diabetes, and hyperlipidemia based on Model 2. Additionally, restricted cubic splines (RCSs) were used to describe the dose-response relationship between PA and sarcopenia incidence through the recommended 3–5 nodes. Fully adjusted RCS models were used to examine the nonlinear relationship between the intensity and occupational nature of PA and sarcopenia [40].

Two methods of sensitivity analysis were used to test the robustness of the results. First, missing covariate data were imputed by using a mode imputation. The results and figures obtained from imputed datasets are presented. Second, the participants were divided into three categories according to their frequency of drinking (never, every day, and not every day), and the variable was introduced into the regression model with other covariates instead of the dichotomous variable (not drinking, drinking).

All the statistical analyses were performed using SPSS 25.0 (IBM, USA) and STATA 16.0 (Stata Corp LLC, College Station, TX, USA). Figures were drawn by using R version 4.3.1 and GraphPad Prism 9 software (GraphPad Software Inc., La Jolla, CA, USA). A two-sided *p* value < 0.05 was considered to indicate statistical significance.

Results

The baseline characteristics of the participants are shown in Table 1. A total of 770 (8.7%) of the participants were living with sarcopenia. The mean age of the participants was 39.2 years, and nearly half of the participants (50.4%) were female. A total of 36.8% of the participants were non-Hispanic white. The majority of the participants (83.5%) had a high school education or above, and more than half were married (59.7%). More than half of the participants had hyperlipidemia (68.7%); however, most

Table 1 Baseline characteristics of the participants in NHANES 2011–2018

Characteristic	Total (n=8802)	Non-sarco- penia (n=8032)	Sarcope- nia (n=770)	p-value
Age (Mean ± SD)	39.2 ± 11.5	38.8 ± 11.5	43.6 ± 11.3	<0.001**
PHQ-9 score (Mean ± SD)	3.16 ± 4.30	3.11 ± 4.24	3.75 ± 4.85	<0.001**
Sex				0.717
Male	4369 (49.64)	3982 (49.58)	387 (50.26)	
Female	4433 (50.36)	4050 (50.42)	383 (49.74)	
Race				<0.001**
Mexican American	1294 (14.70)	1033 (12.86)	261 (33.90)	
Non-Hispanic white	3236 (36.76)	3017 (37.56)	219 (28.44)	
Non-Hispanic black	1796 (20.40)	1746 (21.74)	50 (6.49)	
Other	2476 (28.13)	2236 (27.84)	240 (31.17)	
Marital status				0.023*
Married	5253 (59.68)	4764 (59.31)	489 (63.51)	
Unmarried	3549 (40.32)	3268 (40.69)	281 (36.49)	
Education level				<0.001**
Below high school	1456 (16.54)	1231 (15.33)	225 (29.22)	
High school or above	7346 (83.46)	6801 (84.67)	545 (70.78)	
Poverty income ratio				<0.001**
Poor	1913 (21.73)	1699 (21.15)	214 (27.79)	
Not poor	6889 (78.27)	6333 (78.85)	556 (72.21)	
Nutritional status				<0.001**
Normal	2812 (31.95)	2749 (34.23)	63 (8.18)	
Overweight	2719 (30.89)	2540 (31.62)	179 (23.25)	
Obesity	3271 (37.16)	2743 (34.15)	528 (68.57)	
Smoking				0.115
Yes	3503 (39.80)	3217 (40.05)	286 (37.14)	
No	5299 (60.20)	4815 (59.95)	484 (62.86)	
Alcohol use				<0.001**
Yes	6946 (78.91)	6405 (79.74)	541 (70.26)	
No	1856 (21.09)	1627 (20.26)	229 (29.74)	
Diabetes				<0.001**
Yes	982 (11.16)	803 (10.00)	179 (23.25)	
No	7820 (88.84)	7229 (90.00)	591 (76.75)	
Hypertension				<0.001**
Yes	2488 (28.27)	2179 (27.13)	309 (40.13)	
No	6314 (71.73)	5853 (72.87)	461 (59.87)	
Hyperlipidemia				<0.001**
Yes	6046 (68.69)	5419 (67.47)	627 (81.43)	
No	2756 (31.31)	2613 (32.53)	143 (18.57)	

* $P < 0.05$, ** $p < 0.01$

of the participants were living without diabetes (88.8%) or hypertension (71.7%).

Figure 2 shows the differences in METs across different characteristics between the participants with and without sarcopenia. Total METs, METs for moderate-intensity

PA, and METs for vigorous-intensity PA were significantly greater in the non-sarcopenia group than in the sarcopenia group (Fig. 2A–C). After stratification by occupation, the comparison of the METs for recreational PAs between sarcopenia patients and non-sarcopenia patients was similar to the results without stratification (Fig. 2G–I). However, there were no significant differences in total METs for occupational PA or METs for occupational vigorous-intensity PA across the groups of participants with and without sarcopenia (Fig. 2D and F).

The cross-sectional associations between different intensities of PA (total, moderate, and vigorous) and sarcopenia are presented in Table 2. In the fully adjusted model, compared to participants without PAs (METs were 0), participants in the “low”, “moderate”, and “high” total METs groups had lower risks of sarcopenia (low: OR=0.730, 95% CI: 0.583; 0.915, p -value=0.006; moderate: OR=0.643, 95% CI: 0.485; 0.853, p -value=0.002; high: OR=0.560; 95% CI: 0.457; 0.687, p -value<0.001). The stratified analysis of the associations between moderate- or vigorous-intensity METs and sarcopenia also yielded results similar to those of the non-stratified analysis.

Table 3 shows the association between the intensity of occupational and recreational PA and sarcopenia. According to the fully adjusted model, the participants in the “high” total METs group had a significantly lower risk of sarcopenia (high: OR=0.789; 95% CI=0.653; 0.954; p -value=0.014). Similarly, “high” METs for moderate-intensity occupational PA were associated with a lower risk of sarcopenia (high: OR=0.797, 95% CI: 0.649; 0.980, p -value=0.031); however, higher METs for vigorous-intensity occupational PA were not associated with sarcopenia compared to the reference group. A higher intensity of recreational PA was associated with a lower risk of sarcopenia, i.e., the participants who were labeled with “low”, “moderate” and “high” total METs for recreational PA had a decreased risk of sarcopenia (low: OR=0.758, 95% CI: 0.623; 0.921, p -value=0.005; moderate: OR=0.725, 95% CI: 0.553; 0.950, p -value=0.020; high: OR=0.444, 95% CI: 0.327; 0.604, p -value<0.001) compared to the reference group.

Figure 3 shows the dose-response relationships between different categories of PA intensity and sarcopenia. The relationships between total METs, vigorous-intensity METs and sarcopenia showed a nonlinear U-shaped trend (Fig. 3A and C). However, the relationship between moderate-intensity METs and sarcopenia revealed an L-shaped nonlinear trend. After stratifying PA by occupational and recreational type, the relationships between occupational moderate- and vigorous-intensity PA and sarcopenia showed shapes similar to those of the unstratified relationship, whereas the nonlinear trend became nonsignificant (Fig. 3E and F). In

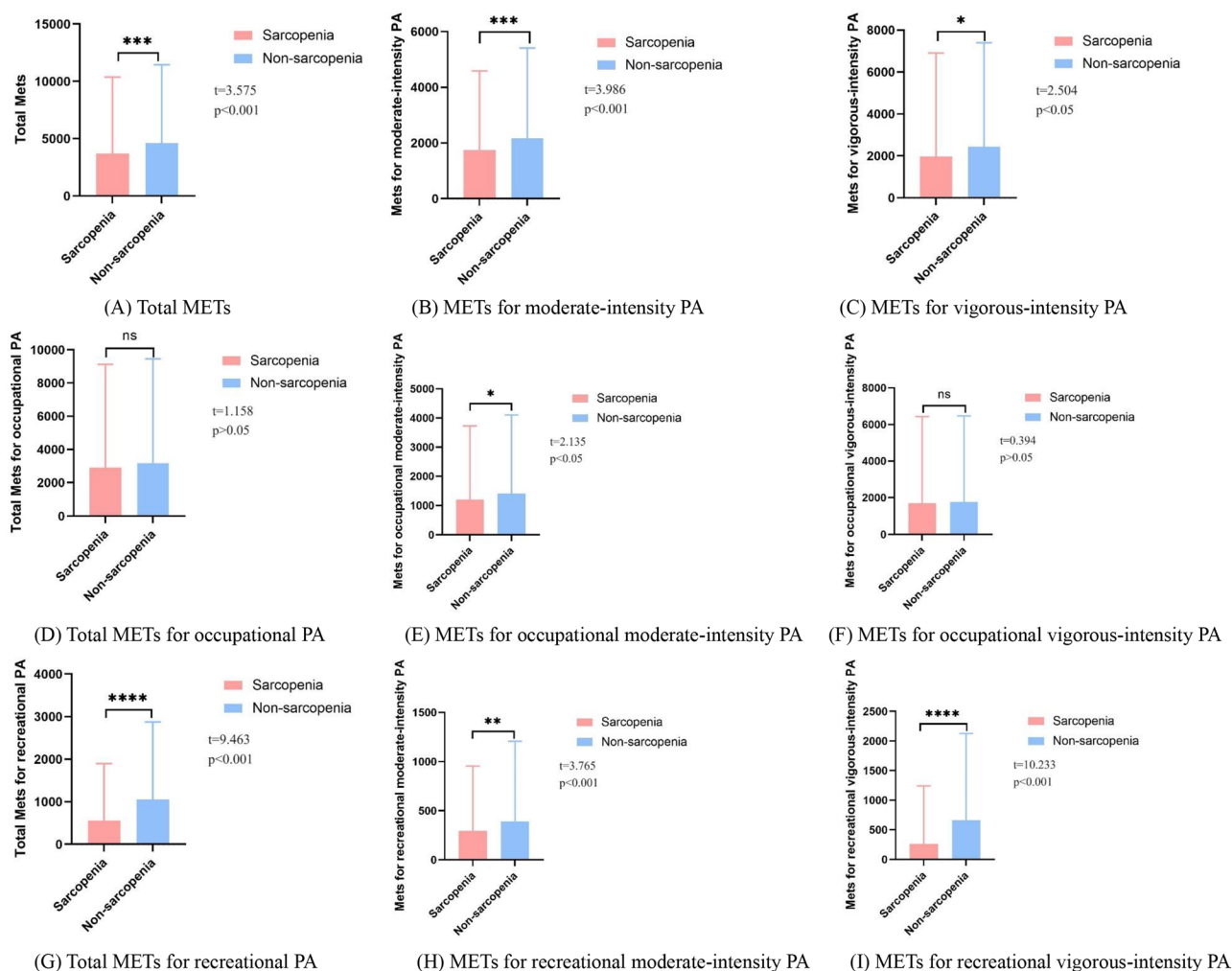


Fig. 2 METs comparison between participant groups with or without sarcopenia

contrast, the relationships between the total METs for recreational PA and the METs for recreational vigorous-intensity PA showed similar shapes and nonlinear trends to the unstratified results (Fig. 3G and I).

Forest plots were drawn to show the associations between different categories of PA intensity (i.e., total METs, occupational PA, recreational PA) and sarcopenia stratified by sex, ethnicity, education, marital status, and poverty status (Fig. 4). Stratified analysis revealed that male, education level of high school and above, and marital status were more likely to reduce the risk of sarcopenia at higher total METs for PA, as shown in Fig. 4A. According to the analysis of the association between occupational PA and sarcopenia, higher total METs for occupational PA did not reduce the risk of sarcopenia in most of the subgroups (Fig. 4B). However, higher total METs for recreational PA significantly decrease the risk of sarcopenia in many subgroups (Fig. 4C). For example, married participants who were categorized as having “low,” “moderate” or “high” METs for recreational PA had

a lower risk of sarcopenia than did the reference group (low: OR=0.692, 95% CI: 0.540; 0.888, p -value<0.05; moderate: OR=0.664, 95% CI: 0.470; 0.937, p -value<0.05; high: OR=0.428, 95% CI: 0.287; 0.637, p -value<0.05).

The results of the sensitivity analysis obtained from the imputed data were consistent with the results of the main analysis. The cross-sectional associations between different intensities of PA (total, moderate, and vigorous) and sarcopenia are presented in Supplemental Table 1. Compared to the participants without any PAs (METs were 0), the participants in the “low,” “moderate,” and “high” total MET groups had lower risks of sarcopenia (low: OR=0.730, 95% CI: 0.583; 0.915, p -value=0.006; moderate: OR=0.643, 95% CI: 0.485; 0.853, p -value=0.002; high: OR=0.560, 95% CI: 0.457; 0.687; p -value<0.001). Supplemental Table 2 shows the association between the intensity of occupational and recreational PA and sarcopenia. In the fully adjusted model, the participants in the “high” total occupational MET group had a significantly lower risk of sarcopenia (high: OR=0.789, 95% CI: 0.653;

Table 2 Association between METs different intensity of PA and Sarcopenia

	Model 1			Model 2			Model 3		
	OR	95% CI	p-value	OR	95% CI	p-value	OR	95% CI	p-value
Total METs									
None	Reference			Reference			Reference		
Low (1-1000)	0.609	0.494;0.750	< 0.001	0.617	0.500;0.762	< 0.001	0.730	0.583;0.915	0.006**
Moderate (1001–2000)	0.442	0.340;0.574	< 0.001	0.480	0.369;0.626	< 0.001	0.643	0.485;0.853	0.002**
High (2001-)	0.434	0.361;0.522	< 0.001	0.486	0.402;0.588	< 0.001	0.560	0.457;0.687	< 0.001**
METs for moderate-intensity PA									
None	Reference			Reference			Reference		
Low (1-1000)	0.531	0.438;0.643	< 0.001	0.540	0.445;0.655	< 0.001	0.646	0.525;0.795	< 0.001**
Moderate (1001–2000)	0.627	0.492;0.798	< 0.001	0.684	0.536;0.872	0.002	0.803	0.619;1.042	0.099
High (2001-)	0.537	0.443;0.650	< 0.001	0.584	0.481;0.710	< 0.001	0.638	0.518;0.785	< 0.001**
METs for vigorous-intensity PA									
None	Reference			Reference			Reference		
Low (1-1000)	0.562	0.435;0.728	< 0.001	0.621	0.478;0.806	< 0.001	0.686	0.520;0.905	0.008**
Moderate (1001–2000)	0.368	0.265;0.512	< 0.001	0.412	0.295;0.574	< 0.001	0.477	0.337;0.676	< 0.001**
High (2001-)	0.564	0.467;0.683	< 0.001	0.631	0.517;0.770	< 0.001	0.645	0.522;0.796	< 0.001**

* $p < 0.05$, ** $p < 0.01$

Logistic regression models

Model 1: no covariates were adjusted

Model 2: was adjusted for demographic factors, including sex, and age

Model 3: was adjusted for sex, age, race, education, marital status, poverty-income ratio, nutritional status, smoking, alcohol use, depression, hypertension, diabetes, and hyperlipidemia

0.954; p -value=0.014). For moderate-intensity occupational PA, the “high” group had a lower risk of sarcopenia (high: OR=0.797, 95% CI: 0.649; 0.980; p -value=0.031). Supplemental Fig. 1 shows the dose-response relationships between different categories of PA intensity and sarcopenia severity, which are consistent with the main analysis.

The second method of sensitivity analysis, which included three categories of alcohol consumption (never, every day, and not every day) instead of the dichotomous variable (not drinking, drinking), also showed similar results (Supplemental Table 3). Compared to the participants without any PAs (METs were 0), the participants in the “moderate” and “high” total MET groups had lower risks of sarcopenia (moderate: OR=0.639, 95% CI: 0.464; 0.881, p -value<0.01; high: OR=0.579, 95% CI: 0.460; 0.728, p -value<0.001). Supplemental Table 4 shows the association between the intensity of occupational and recreational PA and sarcopenia. In the fully adjusted model, the participants in the “high” total occupational MET group had a significantly lower risk of sarcopenia (high: OR=0.777, 95% CI: 0.631; 0.957; p -value=0.018). For moderate-intensity occupational PA, the “high” group had a lower risk of sarcopenia (high: OR=0.757, 95% CI: 0.602; 0.952; p -value=0.017).

Discussion

This study investigated the associations between PA intensity, occupational engagement, and sarcopenia among younger adults by using a nationally

representative sample from the USA. The findings of the current study indicated that a greater intensity of PA was associated with a lower risk of sarcopenia, with vigorous and moderate PA exhibiting a similar benefit on sarcopenia. Interestingly, more intensive recreational PA was related to a lower risk of sarcopenia; however, no beneficial influence was found on occupational PA. To the best of our knowledge, this study is the first to investigate the relationships among activity intensity, occupational nature, and sarcopenia among adults. Although the evidence was based on an observational study with a cross-sectional design, it can also provide clues for developing more precise PA recommendations for adults for the early prevention of sarcopenia in adulthood.

Accumulating literature suggests that engaging in more intensive PA may lower the risk of sarcopenia [41–45], which is in line with the findings of the present study, although some studies have drawn inconsistent conclusions, which may be due to different measurements or diagnostic criteria used [46, 47]. PA may increase the levels of hormones such as testosterone [48] and insulin-like growth factor-1 (IGF-1) [49] which promote muscle buildup, inhibit TNF- α and other various molecular mechanisms to promote mitochondrial metabolism, upregulate peroxisome proliferator-activated receptor gamma coactivator 1 α (PGC-1 α) and downregulate Toll-like receptors, which enhances anti-inflammatory and anti-atrophic functions in skeletal muscle [50]. Moreover, PA can also stimulate phosphatidic acid, which may activate protein synthesis and inhibit autophagy to

Table 3 Association between METs of occupational and recreational PA and Sarcopenia

	Model 1			Model 2			Model 3		
	OR	95% CI	p-value	OR	95% CI	p-value	OR	95% CI	p-value
Occupational PA									
Total METs									
None	Reference			Reference			Reference		
Low (1-1000)	0.881	0.688;1.128	0.315	0.882	0.688;1.132	0.325	0.867	0.665;1.131	0.292
Moderate (1001-2000)	0.844	0.600;1.188	0.332	0.853	0.604;1.203	0.363	0.830	0.577;1.193	0.314
High (2001-)	0.805	0.677;0.957	0.014	0.859	0.720;1.026	0.093	0.789	0.653;0.954	0.014*
METs for moderate-intensity PA									
None	Reference			Reference			Reference		
Low (1-1000)	0.855	0.678;1.078	0.185	0.863	0.683;1.089	0.215	0.841	0.656;1.078	0.172
Moderate (1001-2000)	0.744	0.530;1.045	0.088	0.779	0.554;1.097	0.152	0.728	0.508;1.042	0.083
High (2001-)	0.777	0.641;0.941	0.010	0.830	0.684;1.008	0.060	0.797	0.649;0.980	0.031*
METs for vigorous-intensity PA									
None	Reference			Reference			Reference		
Low (1-1000)	1.066	0.752;1.512	0.718	1.027	0.722;1.462	0.882	0.893	0.613;1.301	0.556
Moderate (1001-2000)	0.675	0.404;1.127	0.133	0.687	0.410;1.151	0.153	0.700	0.409;1.200	0.194
High (2001-)	0.913	0.745;1.118	0.378	0.968	0.785;1.192	0.757	0.830	0.664;1.037	0.101
Recreational PA									
Total METs									
None	Reference			Reference			Reference		
Low (1-1000)	0.618	0.516;0.741	< 0.001	0.639	0.533;0.766	< 0.001	0.758	0.623;0.921	0.005**
Moderate (1001-2000)	0.482	0.375;0.621	< 0.001	0.538	0.417;0.695	< 0.001	0.725	0.553;0.950	0.020*
High (2001-)	0.297	0.222;0.396	< 0.001	0.352	0.263;0.472	< 0.001	0.444	0.327;0.604	< 0.001**
METs for moderate-intensity PA									
None	Reference			Reference			Reference		
Low (1-1000)	0.574	0.484;0.681	< 0.001	0.588	0.495;0.698	< 0.001	0.700	0.583;0.841	< 0.001**
Moderate (1001-2000)	0.778	0.574;1.055	0.106	0.831	0.612;1.129	0.237	1.040	0.752;1.438	0.813
High (2001-)	0.501	0.295;0.850	0.010	0.543	0.319;0.924	0.024	0.538	0.308;0.939	0.029*
METs for vigorous-intensity PA									
None	Reference			Reference			Reference		
Low (1-1000)	0.448	0.337;0.596	< 0.001	0.511	0.383;0.682	< 0.001	0.615	0.454;0.834	0.002**
Moderate (1001-2000)	0.320	0.225;0.455	< 0.001	0.364	0.255;0.519	< 0.001	0.429	0.297;0.620	< 0.001**
High (2001-)	0.282	0.192;0.415	< 0.001	0.344	0.233;0.508	< 0.001	0.444	0.296;0.666	< 0.001**

* $p < 0.05$, ** $p < 0.01$

Logistic regression models

Model 1: no covariates were adjusted

Model 2: adjusted for demographic factors, including sex, and age

Model 3: adjusted for sex, age, race, education, marital status, poverty-income ratio, nutritional status, smoking, alcohol use, depression, hypertension, diabetes, and hyperlipidemia

maintain muscle mass [51]. Notably, the risk of sarcopenia decreased if a certain intensity of PA was reached, regardless of whether vigorous or moderate PA was taken; however, it may be difficult for individuals to benefit from extremely intensive PA. The possible reason may be muscle impairment and growth restriction due to overuse of exercise [52].

Regardless of the general benefit of PA, the protective role seems to exist only in recreational activities, which is consistent with previous studies [53, 54]. In contrast, occupation of PAs was not associated with sarcopenia. High labor-intensity PAs may generate physical and mental stress, which may adversely affect muscle function as

a result of low autonomy and inflexible working hours [27], leading individuals to fail to meet their muscle mass maintenance needs [55]. Additionally, individuals who engage in occupational PA may tend to have relatively lower socioeconomic status and be exposed to a wide range of adverse work environments [27], which increases their vulnerability to adverse health outcomes. For example, individuals who have a manual occupation usually have an elevated risk of mortality [56], hospitalization for influenza [57] and cognitive impairment [58]. Further research is needed to determine the associations between PA and sarcopenia among adults with physically demanding work.

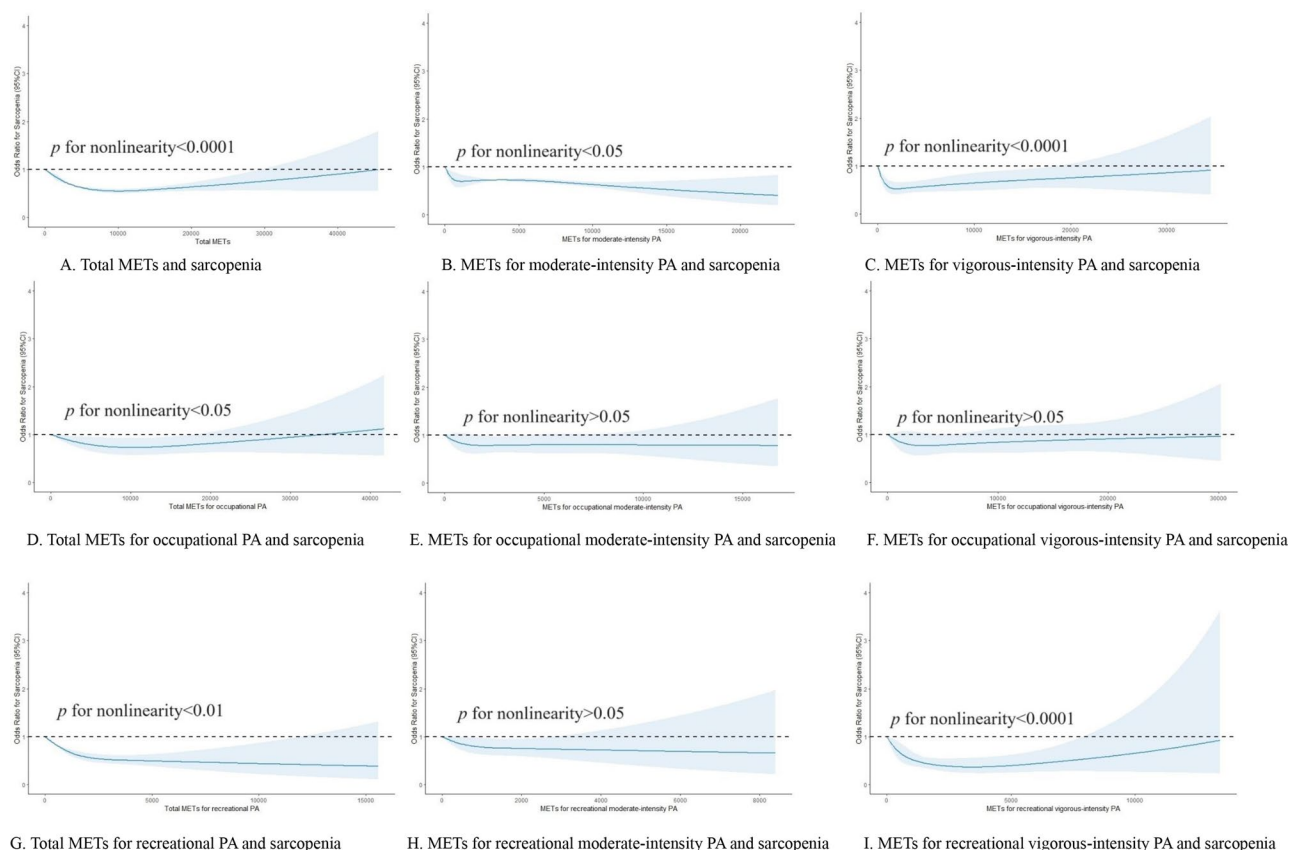


Fig. 3 Dose-response relationship between different categories of PA intensity and sarcopenia

The current study also revealed that some demographic factors may modify the associations between PA and sarcopenia; for instance, males and individuals with relatively high educational levels seemed to benefit more from being physically active, which is consistent with previous studies [59]. Individuals with different sociodemographic characteristics may have different PA engagement patterns; for example, females are more likely to participate in physical activity in their leisure time [55]; however, males with low incomes (who are closely connected to those with low education levels) are predominantly engaged in physically demanding work. The modification effects of some sociodemographic factors make it difficult to interpret the findings, so future studies should enroll individuals with different sociodemographic characteristics to further verify the findings.

This study has the following advantages. First, a nationally representative sample with a large sample size was used for the analysis so that the scientificity and reliability of this study could be guaranteed. Second, sarcopenia was defined by the use of authoritative diagnostic criteria, and limb muscle mass was objectively measured, which was helpful for reducing misclassification bias. Additionally, multiple covariates were included in this study to minimize the interference of confounders. However,

there are still some limitations in this study. First, the cross-sectional study design made it unable to determine any causal relationship between PA intensity, occupational engagement, and sarcopenia. Prospective cohort studies are needed to verify the findings of the present study. Second, information on PA was self-reported, which made it impossible to completely avoid reporting bias. Third, only the ratio of ALM to BMI based on the measurement of muscle mass was considered in the definition of sarcopenia in the present study [8, 60, 61]. Since muscle function data (e.g., gait speed) were not available in the NHANES, more comprehensive criteria, such as the EWGSOP and AWGS, could not be applied in the current study. Future studies should include more measurements and further verify the findings obtained from the current study. Additionally, some factors that may potentially impact our findings could not be fully considered because such data were restricted or missing. Hence, such limitations should be considered in future studies; for example, personal medication and employment type should be included in the future study design. Finally, the samples used in the study were predominantly European and American, so the conclusions might be difficult to generalize to other populations.

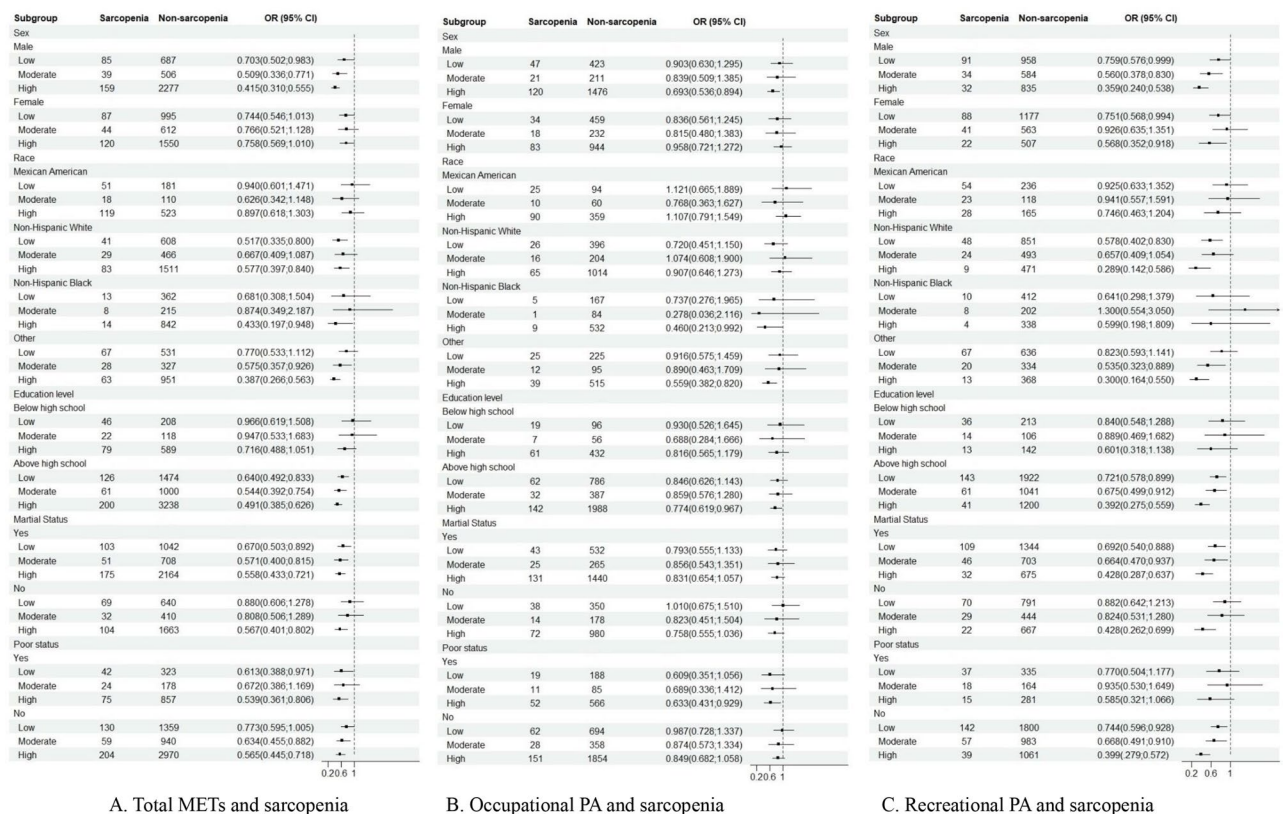


Fig. 4 Forest plots showing the associations between total METs, occupational PA, recreational PA and sarcopenia, stratified by sex, ethnicity, education, marital status, and poverty status

Conclusion

To summarize, the current study concluded that engagement in PA, irrespective of moderate or vigorous PA, is associated with a lower risk of sarcopenia for younger adults. Regardless of whether it is moderate or vigorous in nature, PA contributes positively to a decreased risk of sarcopenia. Moreover, recreational PAs are particularly advantageous compared with occupational PAs, which may not yield the same benefits. Early prevention of sarcopenia is highly important because of the lack of effective treatments to reverse the disease process. Our findings may provide a reference basis for developing more precise prevention strategies for sarcopenia in early adulthood.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-024-20665-9>.

Supplementary Material 1

Author contributions

W Z drafted the manuscript. G S and Y Z framed the concept and designed the study. Data management and cleaning were conducted by W Z, C D, Q W and J Z. X L, W Z performed the data analysis and interpretation. A critical review and check of the manuscript were performed by Y Z, G S and R C. All

the authors meet the criteria for authorship according to their contributions to the manuscript. All the authors contributed to the article and approved the submitted version.

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Data availability

The datasets generated and/or analyzed during the current study are publicly available in the NHANES repository at the following URLs: 2011–2012 NHANES data: <https://www.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2011;2013-2014> NHANES data: <https://www.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2013;2015-2016> NHANES data: <https://www.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2015;2017-2018> NHANES data: <https://www.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2017>.

Declarations

Ethics approval and consent to participate

This study was conducted according to the guideline laid down in the Declaration of Helsinki. All procedures performed in the study involving human participants were approved by the Ethics Review Committee of the National Center for Health Statistics (available at <https://www.cdc.gov/nchs/nhanes/irba98.htm>), and the data were free and publicly available. The information of each participant in the NHANES was collected after providing

informed consent. The NHANES website provides detailed information on ethical declaration and written informed consent.

Consent for publication

Not applicable.

Conflict of interest

None declared. The research was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

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